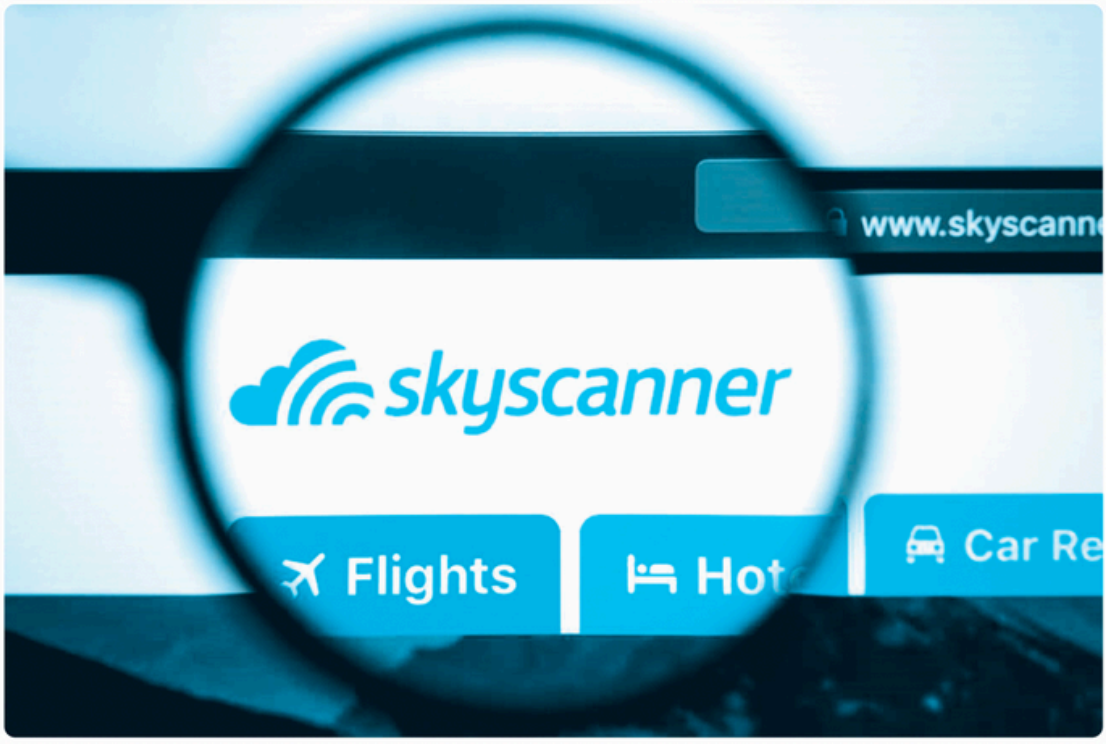
4.8 CASE STUDY

Skyscanner:
The One-Stop Travel Platform

Founded in 2003 by three IT professionals, Gareth Williams, Barry Smith, and Bonamy Grimes, Skyscanner is one of the world's leading travel metasearch platforms. Born as a result of Williams's frustration over a search for cheap flights, Skyscanner started its operations out of his spare bedroom as a site to make traveling easy by simply listing all flight options in one place. A year later, the company opened its first offices in Edinburgh, Scotland. Since then, Skyscanner has become a notable platform for travel information aggregation services. In 2016, Skyscanner was acquired by the Trip.com Group for £1.4 billion, making it the largest acquisition in the travel technology space in Europe. Today, Skyscanner has more than 90 million monthly users.

The company provides a one-stop platform where travelers can plan their flights, hotels, and car hire through aggregation of information from different sources. In its initial stage, Skyscanner's business model was based on its core service of providing flight price aggregation to travelers and receiving commissions for bookings, but it soon added hotel and car hire services and branched into providing travel data insights



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to travel industry players like airports and airlines. Skyscanner's business and revenue models evolved over time to accommodate a two-sided marketplace information aggregation model—providing travel information to both customers and industry players.

To become a leader in the travel data aggregation industry, Skyscanner made three key strategic decisions. First, it expanded its services to other countries like China, Singapore, Australia, and the United States through the launch of different international websites. These websites provided support in over 30 languages, including French, Portuguese, Russian, Chinese, Spanish, Japanese, and German. To support this expansion, Skyscanner secured about £1.5 million in Series A funding from Scottish Equity Partners and another £1 million in 2008.

Second, Skyscanner added hotel and car hire services to its travel planning offerings. The travel booking process is often considered a booking funnel, which means that people book flights first and then hotels, cars, and other ancillary services and products. With the addition of the hotel booking and car hire services, Skyscanner offered its users the convenience of looking for all information on flights, hotels, and car hire services on the same website, without having to navigate several different sites. As Skyscanner's core service was flight information aggregation, it lacked the requisite partnerships to effectively offer hotel and car hire services, so it acquired Spanish hotel comparison business Fogg to leverage its resources to offer hotel and car hire services.

Lastly, Skyscanner launched an app, available for both Android and iOS mobile devices, that combines Skyscanner's three main services into one platform to enable complete trip planning, irrespective of the user's location. Besides providing price comparison of flights, hotels, and car hire services from different providers, the app also furnishes information on available deals and suggests activities at the desired travel destinations. Additionally, the Skyscanner app offers a "recent search" feature to allow users to continue from where they left off on previous searches. The success of the app can be attributed in large part to the acquisition of Distinction, an app development company, and Twizoo, the world's largest social media review and recommendation company.

Skyscanner also launched Travel Insight, a service that derives insights from travelers' flight searches and redirects data to provide information to airlines and airports to optimize prices, plan routes, anticipate market trends, and learn more about travelers. In 2022, Skyscanner created a new data tool to Travel Insight, called Catchment Analysis, to help airports understand why a travel might choose to book travel out of one airport as compared to another. Skyscanner also added a direct booking service via its platform instead of redirecting users to service providers.

While there are similar services in the travel price aggregation and metasearch industry, such as Kayak, Booking.com, Momondo, Trivago, and Expedia, Skyscanner sets itself apart from the competition with cheap prices, a user-friendly website and app, and the two-side marketplace model that provides services to travelers and service providers. Through integration with over 1,200 travel companies, Skyscanner is able to compare options across a large number of websites and provide cheaper prices as well as price deals and special offer alerts to its customers.

It is estimated that Skyscanner has a nearly 4% market share of the world's online booking app business; this is in a market dominated by Booking.com, Hopper, and Airbnb. Skyscanner saw around 5.2 million downloads of its booking app in 2021,

down from a high of 14.9 million in 2018. It is not alone in struggling to regain market share after the Covid-19 pandemic. In 2022, *Which?* magazine published the results of an October 2022 survey involving nearly 3,200 members regarding their experiences researching holidays online in the two-year period prior to the survey. Skyscanner shared the top spot with two competitors—Google Flights (airfares only) and Kayak (both flight prices and hotel prices)—with a customer score of 64%.

Skyscanner has also joined the global transformation in the industry to sustainable travel. It has been a pioneer in showing consumers the carbon emissions of their flights at point of purchase, going beyond its competitors by analyzing aircraft models and calculating carbon dioxide (CO_2) emissions to identify eco-friendly flights, enabling travelers to make an informed decision about their carbon footprint. Initiatives like this make travelers and airlines more conscious about the impact of travel on climate change and encourage them to take steps to reduce their carbon footprints. Through the "greener choices" campaign, Skyscanner has helped over 10 million customers select low-emissions flights.

Behind the technology used by Skyscanner are hundreds of engineers working to develop fast, affordable, and user-friendly software. To sustain its dominance in its field, Skyscanner needs to maintain the most modern technology infrastructure. The company thus evolved its core technology from Microsoft Excel to a web-based platform and mobile application.

Skyscanner's technology runs on Amazon Web Services (AWS), an on-demand cloud computing infrastructure that provides a host of tools, technologies, and support services. AWS optimizes the performance of Skyscanner's platforms by automatically reducing and expanding infrastructure requirements based on Internet traffic. With support from AWS, Skyscanner has migrated its entire infrastructure to the cloud to ensure better performance and cost optimization. It uses cloud cost intelligence platform CloudZero to help it further optimize cloud cost expenses. As such, Skyscanner is able to provide services on a cost-effective basis to more than 90 million customers across the world every month.

Skyscanner also switched to GitHub enterprise services in 2018 for its technology development needs. As Skyscanner added new services and functionalities, its codebase (a collection of source code) grew as well. Hence, there was a need for a robust development platform that supports faster source code deployment and version control that involves support for concurrent work on the same source code as well as easy maintenance. As GitHub's enterprise platform offers increased development speed, automated workflows of software development, and open collaboration between different software programmers, it aligns perfectly with Skyscanner's requirements. GitHub has become the development platform used by every Skyscanner engineer, who makes about 100 to 200 source code deployments every day. A few months after switching to GitHub, Skyscanner reported low operational overheads as engineers did not need to spend a lot of time maintaining the development platform while dealing with source code version control. More time is now dedicated to developing business logic and improving their service offerings.

Moreover, through GitHub, Skyscanner has been able to integrate with and use third-party apps like Snyk and Check. Snyk, for instance, scans login tokens to detect and prevent unauthorized access to its source code. Through GitHub, Skyscanner is also able

SOURCES: "How Skyscanner Embedded a Team Metrics Culture for Continuous Improvement," by Ramsay Ashby and Laz Allen, Infoq.com, March 8, 2023; Best and Worst Travel Comparison Websites, and the Cheapest," *Which?*, which.co.uk, February 8, 2023; "Travel App Revenue and Usage Statistics (2023)," by David Curry, *Business of Apps*, businessofapps.com, January 9, 2023; "Skyscanner Wants to Help Airports Understand 'Passenger Leakage'," Phocuswire.com, October 18, 2022; "How Skyscanner Decentralized Cloud Cost to Their Engineering Teams," Cloudzero.com, July 13, 2022; "Customer Story—Skyscanner & GitHub," Github.com, accessed October 18, 2020; "Telling the Skyscanner Story," Skyscanner.net, accessed October 15, 2020; "Why Skyscanner?" Skyscanner.net, accessed October 10, 2020; "Skyscanner Case Study—AWS Support," Aws.amazon.com, accessed October 10, 2020; "Go Compare Elsewhere: Which? Reveals Best & Worst Travel Comparison Sites," Press.which.co.uk, December 2, 2019; "Skyscanner: Globalising a Business Model," by Richard Raymond Smith and Adina Wong, *Asian Management Insights*, 4(1), 48–52, Ink.library.smu.edu.sg, 2017; "The Spreadsheet Maker Who Created Scotland's Flying Unicorn," by Mure Dickie and Madhumita Murgia, *Financial Times*, November 25, 2016; "Chinese Online Travel Company Buys Skyscanner for £1.4bn," by Peter Wells, *Financial Times*, November 24, 2016.

to access and use new industry technologies and tools to build its solutions, enabling it to stay ahead of the competition. As such, GitHub has become a critical component of Skyscanner's overall infrastructure, providing an integrated and low-maintenance home for all of Skyscanner's code. For now, Skyscanner hopes to keep improving its core technologies to offer better value to its customers. In 2023, Skyscanner embarked on a project to strengthen its use of metrics to improve the pace and quality of its software development efforts, focusing primarily on culture change within the organization rather than the use of specific tools.

Case Study Questions

1. Why did Skyscanner adopt Amazon Web Services for its information technology infrastructure?
2. What strategic decisions enabled Skyscanner to become a leader in its industry?
3. What are the primary benefits of Skyscanner's mobile app?
4. How did Skyscanner ensure that users could use its services in different countries?
5. How has Skyscanner managed to outperform its competitors?

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